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THE HISTORY OF THE IAA SETI PERMANENT COMMITTEE – 1980 to 1989

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This paper is the second in a series outlining the history of the International Academy of Astronautics (IAA) CETI (communication with extraterrestrial intelligence) Committee, which is now known as the IAA SETI (search for extraterrestrial intelligence) Permanent Committee. The IAA established the SETI Committee in 1974 in response to growing awareness that humanity might detect intelligent extraterrestrial life. The IAA SETI Committee is charged with an extraordinarily broad mandate to examine all aspects of the search for extraterrestrial intelligent life including international issues, astrophysical and astronomical observations, biochemistry, exoplanets, complex life and evolution, planetary space missions, SETI search strategies, and the societal, legal, and political impact of a verified detection. The IAA SETI Committee, now in its 45th year, is the only international body that meets annually to organize symposia, select papers for publication, collaborate with other organizations, and exchange information on SETI topics. Between 1980 and 1989, the SETI Committee addressed several areas of interest, including, but not limited to, SETI search technical parameters and results, SETI search strategies, radio frequency interference, and SETI international policy initiatives. Following up on its session in 1977 with the United Nations Committee on the Peaceful Uses of Outer Space, the SETI Committee studied the legal, political, and social implications of the detection of an ETI signal and prepared a draft 'Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence.' The document was approved by the IAA Board of Trustees and the Board of Directors of the International Institute of Space Law in 1989. During the 1980s, approximately thirty SETI search programs were underway or completed, a large-scale NASA SETI program was under development, the SETI Institute in California, U.S.A. was established, and the search for extrasolar planets intensified. Throughout the 1980s, the IAA SETI Committee continued its work on the technical challenges of SETI and all other aspects of the search for intelligent extraterrestrial life in the universe.

I. INTRODUCTION

A paper describing the founding of and early years of the IAA SETI Committee* was presented at the 66th IAC in Jerusalem, Israel, in 2015 (1).

IAA SETI Committee Background

After the Second World War, the unlikely triumvirate of Germany, Britain, and France organized the first international meeting for members of rocket societies and other groups to discuss non-military space research, space travel, and related topics. The first International Astronautical Congress (IAC) was held in Paris in 1950. The International Astronautical Federation (IAF) was subsequently established to foster dialogue between scientists and support international cooperation for the non-military use of space, including the development of astronautics and interplanetary flight. The IAF continued the annual International

Astronautical Congresses as the IAF's international meeting venue and encouraged a diverse array of space-related topics, including space law. A lawyer, Andrew Haley, presented the first SETI-centric paper at an IAC in 1956. The paper examined how to translate human law onto relations with extraterrestrial intelligence (2).

By 1957 humanity entered the space age, with the launch of Sputnik. In 1960 the IAF, which was a body of member-societies, approved the formation of an international organization of distinguished individuals in the aeronautics field; the International Academy of Astronautics. The International Institute of Space Law (IISL) was formed at the same time. The IAA and the IISL would, under the umbrella of the IAF, hold joint symposia and meetings at the annual International Astronautical Congresses. This structure remains in place today.

The IAA provides a non-governmental forum through which members can contribute to advances in all aspects of the peaceful exploration of space, including space policy, law, economics and space and society.

In 1961 the first focused search for signals from intelligent extraterrestrial civilizations took place in

* In this paper, the SETI Committee refers to the IAA SETI Permanent Committee and the IAA CETI Committee (as it was known until 1987). Technically, the SETI Committee is a Permanent Committee under Commission 1 (Space Physical Sciences) of the International Academy of Astronautics.

April-May, 1960 at Green Bank, West Virginia. Rudolf Pešek, a member of the new IAA, first introduced the topic of SETI to his IAA colleagues in 1965. In 1972 the first IAA CETI symposium took place during the 23rd IAC in Vienna.

In 1974 the IAA established the SETI Committee (then called the CETI Committee), with Rudolf Pešek appointed as Chair. Of particular importance was the view of the new committee that papers, talks, and activities should address both the technological and sociological aspects of the search for intelligent extraterrestrial life in the universe.

The mandate of the SETI Committee, as stated by the International Academy of Astronautics remains as follows:

1. Examine all aspects of possible future contact with extraterrestrial civilizations, with special reference to international issues and activities involving the underlying science of the study of life in the universe including:
 - relevant astrophysical and astronomical phenomena
 - the nature and distribution of biogenic elements and compounds
 - the formation of life-bearing planets
 - the origin and evolution of biological systems
 - complex life and intelligence
 - the emergence of technological civilizations
2. The SETI Committee also looks at associated space missions and technology including:
 - planetary science missions designed to search for evidence of extraterrestrial life
 - the astronomical search for extrasolar planets
 - the recovery of possible Martian microfossils on the Earth
3. The science and technology of SETI includes:
 - the development of search strategies and the underlying intellectual foundation
 - the design, development, construction and operation of telescopes and signal detection systems, studies of their future location in space or on the lunar farside
 - rejection of radio frequency interference
 - the archiving of SETI data, and analysis of search results.

And if that was not enough, the IAA also states that the SETI Committee will continue to lead in discussions of the implications of detecting extraterrestrial signals. This includes the areas of philosophy, historical

analogs, anthropology, legal, political and institutional issues, sociology, psychology, theology, media, and the education system.

Finally, the SETI Committee organizes annual SETI sessions/symposia as part of the International Astronautical Congress (IAC), selects peer-reviewed papers of interest for publication in the scientific journal *Acta Astronautica*, works cooperatively with other groups to co-sponsor conferences, meets annually to discuss SETI matters and establishes taskforces and workgroups to grapple with issues of importance. The SETI Committee is to plan for the detection of intelligent extraterrestrial life and consider the aspects and implications of a potential reply from earth.

The SETI Committee is unique in that it remains the sole long-term non-governmental international body to meet annually and to present papers, talks and examine all aspects of the search for intelligent extraterrestrial life in the universe. The value of the institutional setting for the SETI Committee within the IAA may be evaluated from time to time; this paper and its predecessor may provide context for those discussions.

II. THE DAWN OF THE 1980s

By the end of a tumultuous 1979, the Soviet Union was the dominant space-faring nation, the American space shuttle program was in advanced development, and a cold war was brewing. The Cosmos TV series hosted by Carl Sagan was popular, and the Planetary Society was founded.

The dawn of the 1980s featured the discovery of exotic life surrounding hydrothermal vents in the deep ocean and the identification of the Cretaceous/Tertiary extinction event, leading to new ideas and advances in the understanding of exobiology and the impact of catastrophic events on life. Conversely, some scientists argued that life on earth might be unique, given its complexity and unlikely series of circumstances conducive to evolving an intelligent organism.

In 1980 the 9th IAA International Review Meeting on SETI took place at the 31st IAC in Tokyo, Japan. The thirteen papers presented included a review of unconventional and non-microwave methods of CETI and SETI (3) and the concept of using the sun's gravity lens for SETI (4). Ivan Almar and Jill Tarter, two individuals who significantly contributed to the SETI Committee for many decades, are listed as co-authors on several papers presented in Tokyo. John Billingham, Co-Chair of the SETI Committee, M. Subotowicz, and Stark Draper, President of the IAA, were the three

attendees at the SETI Committee meeting. They discussed the effort to solicit papers from western and eastern bloc countries and the need to invite new people to be members of the SETI Committee. It was decided at the meeting to seek papers on the following topics for the next IAC to be held in Rome, 1981:

- Recent advances in chemical evolution and the origin and early evolution of life
- The influence of astrophysical, stellar and stellar system events on the pathways and rates of evolution
- SETI technology
- Interferometry in SETI
- Legal and international aspects of SETI including the problem of radiofrequency interference
- The impact of SETI in terms of scientific, psychological, sociological and cultural implications
- New developments in the field of extrasolar planetary detection

The first launch of the space shuttle Colombia in April 1981 boosted global space exploration efforts. The 10th IAA SETI Review Session 1981 took place during the 32nd IAC in Rome, Italy. Rudolf Pešek provided an overview of the first ten years of CETI Committee review meetings. An American diplomat and policy expert, Michael Michaud, presented a paper on “A Grand Strategy for the Species.” (5) Four SETI Committee members and three guests attended the annual SETI Committee meeting; John Billingham, C. Clemenson, V. Gogosov, M. Subotowicz, W. Hilton, J. Tarter, and Stark Draper (IAA President). It was noted during the SETI Committee meeting that only ten members belonged, and invitations should go out to eleven new prospective members. The topic of media interest in a verified or unverified detection of an extraterrestrial signal was brought up to committee members so they could consider their potential response in advance.

In 1982 the 11th SETI Review Meeting at the 33rd IAC in Paris, France, offered a robust selection of papers, including descriptions of SETI searches undertaken at the Nançay (6), Westerbork (by Seth Shostak, a future Chair of the SETI Committee) (7), and Arecibo (8) radio telescopes, SETI technology, SETI ideas, and sociological and evolutionary aspects of SETI. By 1982 the number of SETI Committee members had increased to twenty:

John Billingham	U.S.A. (Co-Chair)
R. Pešek	Czechoslovakia (Co-Chair)
Ronald Bracewell	U.S.A.

Carl Clemenson	Sweden
Robert Dixon	U.S.A.
Robert Edelson	U.S.A.
L.M. Gindilis	U.S.S.R.
V.V. Gogosov	U.S.S.R.
W.F. Hilton	England
N.S. Kardashev	U.S.S.R.
Michael Klein	U.S.A.
A. Lawton	U.K.
G. Marx	Hungary
Masaki Morimoto	Japan
Zbigniew Paprotny	Poland
Cyril Ponnamperna	U.S.A.
M. Subotowicz	Poland
Jill Tarter	U.S.A.
V.S. Troitskij	U.S.S.R.
R. Weilebinski	Federal Republic of Germany

During the 1982 SETI Committee meeting, John Billingham stressed that topics to be presented at the SETI Review Sessions should also address non-technical areas, including exobiology and the sociological impact of SETI. The SETI Committee also discussed a U.S.S.R. and Estonian Academies of Science SETI meeting held in Tallinn, Estonia, 1981. Michael Papagiannis, President of the new Commission 51 of the International Astronomical Union, attended the meeting as a guest.

In 1982 the United Nations held a meeting on the Peaceful Uses of Outer Space (UNISPACE 82). John Billingham, the IAA SETI Committee co-chair, was asked by the IAF to prepare a background paper on SETI for the delegates to the UNISPACE 82 conference. The paper was presented by the IAF to COSPAR, the Committee on Space Research. Despite the submission of the background paper on SETI, the topic was not included in the Vienna discussions.

The IAA SETI Committee was also asked to prepare a brief background paper on the technology of SETI for the direct IAF input to the United Nations.

The 34th IAC was held in Budapest, Hungary, in 1983. The 12th IAA SETI Review Meeting Sessions featured several talks on the origin of life and early evolution. Of particular interest was the paper by J. Corliss about the recently discovered clusters of life forms thriving around deep ocean hydrothermal vents.

Continued Issue of Radio Frequency Interference

One of the first orders of business for the IAA SETI Committee, when it was founded, was to work towards the protection of specific wavelengths of the radio spectrum to allow for the search of extraterrestrial

signals. The effort peaked with a request by the IAA and IAF to the International Telecommunications Union for consideration at the World Administrative Radio Conference in Geneva in 1979. Still, no space in the spectrum was allocated for SETI activities. By 1980, the SETI Committee began to speculate that SETI activities may need to be space-based to mitigate the interference problem. The issue continued to be a priority for the SETI Committee throughout the 1980s.

III. 1984 – THE FOUNDING OF THE SETI INSTITUTE AND COLLABORATION WITH THE IAU

By the mid-1980s, several SETI searches were underway. Most focused on the microwave and optical parts of the spectrum, but several searches targeted manifestations of extraterrestrial technology including Dyson spheres, spectral tritium lines as evidence of stellar engineering, linear gamma-ray bursts from starships, radar reflections from solar system artifacts and examination of earth's Lagrange points for extraterrestrial probes.

The SETI Committee continued to foster partnerships and cooperation with other like-minded space organizations, including the International Astronomical Union (IAU). The IAU established a commission devoted to the search for extraterrestrial life. In the spirit of encouraging international cooperation, Rudolf Pešek and John Billingham, as Co-Chairs of the IAA SETI Committee, assisted the new IAU Commission 51. Michael Papagiannis was appointed an official representative of the IAU to the IAF and subsequently accepted membership in the IAA SETI Committee. In 1984 the first Bioastronomy Commission Symposium, including talks on SETI related topics, took place in Boston in 1984.[†]

NASA and its partner institutions continued hosting studies, symposia, colloquia, workshops, conferences, and study groups on SETI throughout the 1980s as NASA and partners continued to advance work on the research, design, and implementation of a significant microwave SETI project.[‡] The need to maximize

[†] The IAU Bioastronomy Conferences sponsored by Commission 51 of the IAU were triennial events. The last Bioastronomy Conference was in San Juan, Puerto Rico in 2007. Subsequently, Commission 51 combined the Bioastronomy conference series with the Origins of Life Conferences, and SETI related talks were phased out.

[‡] For a thorough description of this period, see the 2004 book *"The Living Universe – NASA and the*

available funds for SETI work was the impetus for the founding of the SETI Institute in 1984 as a non-profit organization. From its founding 35 years ago to the present, the SETI Institute provides a high level of in-kind support for IAA SETI Committee logistics, administration, and activities.

The 13th IAA SETI Committee Review meeting took place in Lausanne, Switzerland, in 1984 during the 35th IAC. Michael Papagiannis described the proceedings of the first IAU symposium on extraterrestrial life (8) and an anthropologist, Ben Finney, presented on the topic of interstellar migration (9).

In keeping with the multidisciplinary mandate of the SETI Committee, it was suggested that COSPAR be invited to co-sponsor the CETI I session at the 1985 IAC to be held in Sweden. The session would focus on the origin of life, chemical evolution, and Precambrian geology. John Billingham brought up the possibility of a joint session with IISL on the legal, political, and social implications of the detection of extraterrestrial intelligent signals.

The 14th IAA CETI Review Session during the 36th IAC in Stockholm presented a broad array of papers addressing SETI topics, including the previously suggested emphasis on the origin of life and evolution, in addition to the more technical talks. A representative of COSPAR, H.P. Klein, presented a talk.

The SETI Committee meeting in Stockholm was attended by John Billingham, Ivan Almár, W. Hilton, M. Klein, A. Martin, M. Papagiannis, M. Subotowicz, and J. Tarter. The recent lack of Soviet participation in the SETI sessions was noted. The committee again discussed the proposal of a joint session with the IISL on the legal, political, and social implications of the detection of extraterrestrial intelligence. Cooperation with other groups continued, with Michael Papagiannis announcing that a SETI session co-sponsored by Commissions 40 and 51 of the IAU would take place during the IAU General Assembly in New Delhi, in November 1985. Jill Tarter reported on a SETI-focused meeting to be held in Sri Lanka, 1985 and sponsored by The Arthur Clarke Center, the Institute for Fundamental Studies, and the IAA/IAF. A Five-Year Plan for the IAA SETI Committee was prepared to outline meetings and cooperative activities planned or underway.

IV. 1986 – A PIVOTAL YEAR

Development of Astrobiology" by Steven Dick and James Strick.

The tragic loss of the Space Shuttle Challenger and an underwhelming view of a faint Halley's Comet marked the beginning of 1986.

The 15th SETI Review Meeting at the 37th IAC in Innsbruck, 1986, featured a session devoted to legal, political, and social implications of the detection of intelligent extraterrestrial life. This session was of key importance to the SETI Committee, since it brought forward the notion of the SETI Committee developing a code of conduct of sorts in advance of detecting evidence of intelligent extraterrestrial life. The SETI Committee was also interested in the concept of a potential response from earth. Who would prepare a reply, and what should be in it?

The number of SETI Committee members by 1986 had increased to twenty-five as follows:

Member	Country	Joined
Pešek, R.	CSSR	1974
Billingham, J.	U.S.A.	1981
Almár, I.	Hungary	1983
Biraud, F.	France	1983
Bracewell, R.N.	U.S.A.	1981
Clemedson, C.J.	Sweden	1975
Dixon, R.S.	U.S.A.	1981
Gindilis, L.M.	U.S.S.R.	1981
Gulkis, S.	U.S.A.	1985
Hilton, W.F.	U.K.	1981
Kardashev, N.S.	U.S.S.R.	1977
Klein, M.J.	U.S.A.	1981
Lawton, A.T.	U.K.	1975
Martin, A.R.	U.K.	1985
Marx, G.	Hungary	1975
Morimoto, M.	Japan	1981
Pal, Y.	India	1985
Papagiannis, M.D.	U.S.A.	1983
Paprotny, Z.	Poland	1981
Ponnamperuma, C.	U.S.A.	1975
Radhakrishnan, V.	India	1985
Subotowicz, M.	Poland	1975
Tarter, J.C.	U.S.A.	1981
Troitsky, V.S.	U.S.S.R.	1981
Weilebinski, R.	F.R. Germany	1981

Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence

The SETI Committee took on the question of what to do after a signal detection and who speaks for earth as a priority action item. Members of the IAA SETI Committee, with input from many experts, worked for several years on drafting a series of guidelines for individuals or organizations, national or international, engaged in carrying out radio searches for

extraterrestrial intelligence. At the 39th IAC in Bangalore, India, Michael Michaud presented a draft international agreement concerning the detection of extraterrestrial intelligence.[§]

The “Declaration of Principles Regarding Activities Following the Detection of Extraterrestrial Intelligence” was subsequently approved by the Board of Trustees of the IAA, and by the Board of Directors of the IISL in April 1989. The document (“Declaration”) outlines the need for verification and public dissemination of data collection. It states that the discoverer of the detection should have the privilege of making the first public announcement and that relevant radio frequencies should be protected by exercising the extraordinary procedures established within the World Administrative Radio Council of the International Telecommunication Union.

Of particular concern were Items 8 and 9 in the Declaration. Item 8 states that “No response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place. The procedures for such consultations will be the subject of a separate agreement, declaration or arrangement”. Item 9 states that a post-detection committee should be established. The IAA SETI Committee is to maintain a list of willing representatives from the IAA, the IAU, as well as other representatives with the necessary skills.

V. 1987 – 1989 DECLARATION OF PRINCIPLES IS COMPLETED AND ENDORSED BY OTHER GROUPS

In 1987 the focus of the IAA SETI Committee was on further consideration of the legal, political, and social aspects of detection of intelligent extraterrestrial life in addition to technical updates on SETI searches, instrumentation, early life, and other SETI-related topics. The 16th IAA SETI Review Sessions included talks by C. Maccone (a future Chair of the SETI Committee), Jean Heidmann, A. Goodman, A. Tough, C. Devito, and others.

The Declaration draft was presented at the 17th Annual SETI Review Session at the 39th IAC in Bangalore, India, in 1988. In addition to other talks featuring SETI search updates and other SETI-related topics, two talks of interest presented information on SETI searches based in Argentina (9) and the U.S.S.R. (10).

[§] A description of the drafting of the Declaration is provided in the 2007 book “Contact with Alien Civilizations” by Michael Michaud.

The 1989 Tiananmen Square protests in Beijing and martial law declaration by China's government facilitated a change in venue for the 40th IAC from Beijing, China to Torremolinos, Spain, with the result there were several withdrawals of papers. At the start of the annual meeting of the SETI Committee, the Chair, John Billingham announced to attendees that sadly, Rudolph Pešek, the Honorary Chair and founder of the SETI Committee in the IAA, had passed away.

The focus of the SETI Committee in 1989 was the status of the Declaration of Principles and the need to start work on the second document, which would seek an agreement on a potential response from earth. The year 1989 marked the rise of the global internet; Jean Heidmann suggested to committee members that an electronic bulletin board should be set up for all SETI researchers to keep in contact, and this idea was endorsed. As always, the SETI Committee continued its cooperative efforts with other space organizations. John Billingham prepared a brief one page summary on SETI as part of the IAF/IAA input to the 1989 UN report on Highlights of Space Technology. The SETI Committee discussed the idea of holding a special IAA SETI symposium in Washington in association with the World Space Congress in 1992.

In November, 1989 a second special issue of *Acta Astronautica* with selected SETI papers was published.

VIII. SETI COMMITTEE POST 1990 – THE WORK CONTINUES

The third paper in this series on the History of the SETI Committee will review the period 1990 to 1999. During this time the SETI Committee worked vigorously on the contentious issue of transmissions from earth. Significantly, by the end of the 1990s, the first exoplanets were identified, and, suddenly, the concept of extraterrestrial life became closer to reality.

IX. CONCLUSION

The SETI Committee, now in its 45th year, remains an important permanent study group under the auspices of the International Academy of Astronautics. The SETI Committee continues to be committed to its mandate of examining all aspects of the search for intelligent extraterrestrial life in the universe. The actuality of a wide diversity of exoplanets, the continued search for life within our solar system, and the awareness that biological organisms can thrive in extreme environments increase the likelihood of confirming an extraterrestrial biosignature or technosignature. The SETI Committee also considers the legal, policy, and

societal impact of the discovery of intelligent extraterrestrial life. The Declaration of Principles drafted by the SETI Committee in 1989 and its subsequent revisions remain significant.

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REFERENCES

1. Walton, L. and Dumas, S., 2015, The History of the IAA SETI Permanent Committee – The Early Years. Manuscript, IAC-15-A4.2.1, International Astronautical Congress, Jerusalem, Israel.
2. Haley, A.G., 1956, Space Law and Metalaw – A Synoptic View. *Harvard Law Record* 23, Nov 8, 1956
3. Subotowicz, M. and Zaprotny, Z., 1980, Unconventional and Non-Microwave Methods of CETI and SETI. Manuscript, IAA-80-52, 31st International Astronautical Congress, Tokyo, Japan.
4. Eshleman, Von R., 1980, On the Use of the Gravitational Lens of the Sun for Detecting Other Planetary Systems and for Interstellar Communication. IAA-80-59, 31st International Astronautical Congress, Tokyo, Japan.
5. Michaud, M., 1981, A Grand Strategy for the Species. Paper IAA-81-305, 32nd International Astronautical Congress, Rome, Italy.
6. Biraud, Francois, 1982, SETI at the Nancay Radiotelescope. Paper IAA-82-261, 32nd International Astronautical Congress, Paris, France. Unconventional and Non-Microwave Methods of CETI and SETI. Manuscript, IAA-80-52, 31st International Astronautical Congress, Tokyo, Japan.
7. Shostak, G. S. and Tarter, J., "Signal" (Search for Intelligence in the Galactic Nucleus with the Array of the Lowlands). Paper IAA-82-262,

32nd International Astronautical Congress, Paris, France.

8. Tarter, J., Duquet, R.T., Clark, T. and Lesyna, L., Recent SETI Observations at Arecibo. Paper IAA-82-262, 32nd International Astronautical Congress, Paris, France.
9. Colomb, F.R., Martin, M.C., and Lemarchand, G.A., 1988, SETI Observational Program in Argentina. 39nd International Astronautical Congress, Bangalore, India.
10. Gindilis, L.M., Dubinskij, B.A., Rudnitskij, G.M., 1988, SETI Investigations in the U.S.S.R. 39nd International Astronautical Congress, Bangalore, India.